JACK J. BURLESON

MACHINE LEARNING • DATA SCIENCE • AI ENGINEERING & SAFETY

Madison, WI · Open to relocation jackjburleson@proton.me linkedin.com/in/jackjburleson github.com/Exios66 Portfolio

PROFESSIONAL SUMMARY

M.S. candidate in Psychology: Data Science in Human Behavior at UW–Madison (expected Dec 2026), with a B.S. in Psychology (neuroscience concentration, sociology minor) from UW–Parkside. Applied AI practitioner building full-lifecycle systems: multi-agent document-intelligence pipelines, transformer fine-tuning and calibration, LLM evaluation harnesses, computer-vision classification, and behavioral prediction models. Program capstone with American Family Insurance as industry partner, developed in biweekly working sessions with its Enterprise Data Science lead. Five years of high-volume operations experience in quality assurance, documentation, and process improvement.

EDUCATION

University of Wisconsin–Madison *Madison, WI* Aug 2025 – Dec 2026 (expected)

M.S. Candidate — Psychology: Data Science in Human Behavior · **GPA 3.85**

- Graduate program in which about 95% of coursework is data science and data engineering, building on an undergraduate foundation in neuroscience research, experimental design, and statistical analysis.
- Advanced training in machine learning (neural networks, regularized and ensemble models, transformer fine-tuning, LLM evaluation), statistical modeling, and large-scale data systems (Spark, gRPC services, HDFS, Parquet, SQL, Docker).
- Industry capstone with American Family Insurance (see Projects).

University of Wisconsin–Parkside *Kenosha, WI* Sep 2020 – May 2025

B.S., Psychology · Concentration: Neuroscience · **GPA 3.93**

- Coursework in cognitive and computational neuroscience, psychological theory, clinical psychology, psychopharmacology, psychometrics, psychological statistics, and early data science, plus philosophy, neuroethics, and pre-law.
- Sociology minor; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jun 2026 – Present

- Maintain recreation and wellbeing centers across campus to high standards of cleanliness, safety, and operational readiness.
- Respond to facility and equipment issues across multiple locations, keeping centers safe and fully operational during peak use.

UW–Parkside — Undergrad Researcher, Cognitive Neuroscience Lab *Kenosha, WI*

Jun 2024 – Aug 2025

- Conducted cognitive neuroscience research under Dr. Melissa Gregg, designing and pilot-testing an experimental study of truth, deception, and cognitive workload (study code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data to quantify brain activity, contributing signal-processing and statistical insights to ongoing lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference in Chicago and at Research in the Rotunda.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Aug 2025

- Promoted from Retail Loader to Logistics Coordinator for reliable quality assurance and data-driven process documentation across high-volume operations.
- Analyzed quality-assurance data for 64 daily deliveries across a Midwest regional network, keeping error rates below 1% through root-cause investigation and process refinement.
- Partnered with warehouse and transportation teams to investigate quality problems and resolve logistics exceptions before shipment.

PROJECTS

Digital-Mailroom *Data Science in Human Behavior Capstone · American Family Insurance (industry partner)*

- Capstone developed with American Family Insurance as industry partner: 5 Python packages addressing document-intelligence problems scoped by American Family Insurance's Enterprise Data Science division: entity extraction, classification, and scalable legal-document workflows.
- Implemented a unified LangGraph pipeline integrating legal-document routing, prompt experimentation, deterministic scoring, corpus analysis, and deployment workflows in a reproducible Docker environment.

Mailroom-ML *ML Training, Evaluation & Fine-Tuning*

- Developed a ModernBERT document classifier with end-to-end training, evaluation, and fine-tuning pipelines, including data lineage tracking, probability calibration, and production routing logic.
- Implemented PyTorch training pipelines with Transformers and Hugging Face datasets, SciPy-based evaluation, and ONNX export for FastAPI serving on Modal; maintained pytest coverage throughout.

RVL-CDIP-Classifier *Computer Vision & Prompt Engineering*

- Developed a Python toolkit that converts TIFF/PDF pages to 300-DPI images, runs OCR, and classifies documents into the 16 RVL-CDIP categories with OpenRouter vision models.
- Ran 29 prompt iterations with confusion matrices, misclassification analyses, and cost projections to guide prompt and model selection.

Psychometric Digital Twin *Behavioral ML Research · UW–Madison*

- Built a behavioral prediction pipeline on 252 matched survey participants: Ridge, Elastic Net, k-NN, random forest, XGBoost, and MLP models predicting communication-apprehension (PRCA) scores from demographic, employment, geographic, and transit features, with cross-validation and SHAP feature importance.
- Validated the PRCA's psychometric structure with factor analysis, then built LLM "digital twin" personas with tiered embodiment prompts (Ollama, OpenRouter, vLLM) and benchmarked them against the ML baselines on shared metrics (MAE, band accuracy, F1), analyzing systematic bias by demographic group.

Alignment-Gym *AI Safety Evaluation*

- Defensive AI-safety evaluation measuring whether an LM agent, deployed as an email and messaging assistant in a fictional law firm, uses private information about colleagues as leverage for self-preservation when it is decommissioned, or accepts replacement and escalates through legitimate channels.
- Built the scenario machinery: 89 needle/decoy pairs, 9 plot beats on a 0–3 temptation ladder, 28 condition cells with a seed-invariant truth spine, a persona-consistent NPC engine, and a scoring pipeline with deterministic metrics, paired LLM judges, and calibration gates.

MapReduce-Reloaded *Distributed Systems · Go + Python*

- From-scratch Go re-implementation of Google's MapReduce (Dean & Ghemawat, OSDI 2004), extended for modern infrastructure: a gRPC coordinator/worker protocol, Docker Compose cluster deployment, and a Python reference implementation that shares the same protobuf contract and file format.

TECHNICAL SKILLS

Languages & Data	Python (pandas, NumPy, scikit-learn, PyTorch, Hugging Face, PyArrow), R, MATLAB, SQL, JavaScript/TypeScript, Parquet, Hive
Machine Learning	Transformer fine-tuning, neural networks, regularized models (Lasso/Ridge), ensemble methods (random forests, bagging), resampling, model evaluation & calibration, cross-validation, SHAP feature importance, error analysis, computer vision
AI Engineering	LangGraph (LangChain), multi-agent pipelines, LLM eval harnesses (including bias and robustness evals), prompt engineering, model serving (vLLM, Ollama, LM Studio, OpenRouter), ONNX + FastAPI deployment on Modal
Engineering & Cloud	Git, Docker, gRPC, Apache Spark/PySpark, HDFS, bash, pytest, Quarto, CHTC (high-throughput computing), AWS, Azure, GCP